# What are mRNA and AI drug discovery?

How the technologies work and related companies

## Two technologies changing drug development

mRNA is a molecule that conveys instructions for making specific proteins in the body. Research uses this property for infectious-disease vaccines and applications in cancer, rare diseases and other conditions.

## What is AI drug discovery?

Using AI to support the search for disease-related targets and the design and prediction of molecules that may become drug candidates. Combined with researchers’ experiments and clinical trials, it aims to make development more efficient.

## These are different technologies

mRNA is one technological platform for medicines. AI drug discovery is a method supporting R&D across many kinds of medicines, not just mRNA products.
The focus is on how development time, success rates and costs actually change.

# 01 | Expanding mRNA applications

From infectious-disease vaccines toward therapeutic applications

## Moderna

Alongside its infectious-disease vaccine business, Moderna is pursuing R&D applying mRNA to cancer, rare diseases and other conditions. Beyond vaccines, clinical trials are important for verifying efficacy and safety.

## BioNTech

Develops mRNA-based cancer immunotherapies as well as antibody-drug conjugates (ADCs) and other technologies. It aims to expand cancer treatment options by combining multiple technologies.

## Maravai LifeSciences

Supplies reagents and technologies needed for research and manufacturing. Unlike companies selling finished medicines, it supports development of mRNA medicines and other products.
Drug developers and companies supporting research and manufacturing have different business structures and ways of generating revenue.

# 02 | How AI changes drug discovery

Toward R&D linking molecular design and experiments

## Isomorphic Labs

An AI drug discovery company under Alphabet. In May 2026 it announced a US$2.1 billion funding round. Alongside AI model development, it is expanding its capacity to apply the technology to drug design and development.

## Eli Lilly × NVIDIA

In January 2026 they announced a joint AI drug discovery research hub. The companies outlined plans to invest a combined total of up to US$1 billion over five years in personnel, research, computing infrastructure and related areas.

## The R&D cycle

Design candidates with AI → validate through experiments → collect data → improve the model.
Accelerating this cycle is the aim of combining AI with experimental facilities.
Better prediction accuracy does not automatically mean drug approval or success of the development process as a whole.

# 03 | Companies and industry structure

From developing technology to commercializing medicines

## mRNA technology

Moderna / BioNTech
Accumulate design technology and clinical data, aiming for commercialization.

## Drug discovery AI

Isomorphic Labs
Applies molecular design and predictive models in R&D.

## Pharmaceuticals and AI infrastructure

Eli Lilly / NVIDIA
Connect pharmaceutical knowledge and computing infrastructure with R&D.

## Research and manufacturing support

Maravai LifeSciences
Supplies reagents and technologies necessary for development and manufacturing.
Technological progress, clinical trial results and commercialization are distinct stages. Distinguishing each company’s role makes the whole industry easier to understand.